Quantitative analysis of extreme rainfall

In the past two years, Henan, Shaanxi, Hubei and other places of our country have encountered extremely rare heavy rainfall. At the same time, some northern cities have encountered historically rare blizzards. These heavy rains and snowfalls pose a serious threat to the local people about their lives, safety and properties. Take Zhengzhou as an example. From 18:00 on July 18 to 0:00 on the 21st, the city of Zhengzhou saw heavy rainstorms and extreme heavy rains. The cumulative average precipitation was 449 mm. The rainfall at the Zhengzhou station reached 201.9 mm from 16:00 to 17:00 on the 20th, which exceeding the extreme value of rainfall per hour on the land of my country. And Zhengzhou, a city with a population of tens of millions, is undergoing the test of heavy rain. The heavy rain began intermittently on July 17, and by the morning of July 20, the rain suddenly began to increase. By the afternoon of the same day, many communities and roads in Zhengzhou were flooded by rain. The Zhengzhou Meteorological Bureau released information that the average annual rainfall in Zhengzhou is 640.8mm, and the rainfall in these three days is close to or even more than that of previous years. From the perspective of climatology, it is once in a thousand years. The continuous heavy rainfall in Zhengzhou, Henan has caused huge losses to the safety of people's lives and property in Zhengzhou, and has severely affected production and life. As of 12:00 on July 23, according to preliminary statistics, 395,989 people were urgently transferred and resettled, crops affected area of 44,209.73 hectares, direct economic losses were 655 billion yuan, and floods and secondary disasters caused by heavy rains have killed hundreds of people.

According to relevant researchers, in the context of global warming, the amount, intensity, frequency and type of future precipitation in my country will be directly affected. The amount of precipitation is expected to increase by about 10% at the end of this century and the probability of extreme precipitation will increase significantly. Due to the large land area of our country and the combined influence of various types of topography and landforms and other factors, the precipitation characteristics of different cities show different characteristics. Therefore, it is imperative to establish prediction models for cities with different potential extreme precipitation events and quantitative analysis models for their losses.

Attachment 1 shows the daily precipitation observation data of three meteorological stations near Zhengzhou City for the past 70 years. Try to use mathematical modeling to solve the following problems:

Question 1: Carry out the relevant analysis of the annual change characteristics of the precipitation characteristics in Zhengzhou area and screen out a number of years with higher precipitation. At the same time, make a specific quantitative analysis of the inundation event of Zhengzhou in 2021.

Question 2: Can you collect and sort out the precipitation data of more cities in our country for many years, and analyze the trend of precipitation in these cities? (It is necessary to clearly indicate the source of the data and how to obtain it)

Question 3: Can you use the collected urban weather data to predict and analyze

potential cities with extreme rainfall in the future based on different methods, and compare and analyze your prediction effects?

Question 4: Do you think the characteristics of the heavy rainfall in Zhengzhou in July 2021 are the same as those in Shanxi in October 2021? What is the difference in the losses caused.

Question 5: Can you propose a long-term construction plan for cities in the future under extreme precipitation conditions, in which in-depth analysis of typical cities in my country is particularly necessary.

The description of data:

Attachment 1 is the 70-year meteorological observation data of various meteorological stations in Annex 3 of Zhengzhou City.

Your submission should consist of:

- One-page Summary Sheet
- Table of Contents
- One -to Two -page Article
- Your solution of no more than 20 pages, for a maximum of 24 pages with your summary, table of contents, and article

Note: Reference List and any appendices do not count toward the page limit and should appear after your complete solution. You should not make use of unauthorized images and materials whose use is restricted by copyright laws. Ensure you cite the sources for your ideas and the materials used in your report.